



Stevan Pavlović

Backend & Distributed-Systems Engineer · Tech Lead

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SUMMARY

Backend and distributed-systems engineer in Vienna. Eleven years of building event-driven services on Node.js, NestJS, Kafka, PostgreSQL and Redis across fintech, betting, healthtech and fleet logistics. I lead small teams from the first commit to zero-downtime production, and I care about correctness, clear interfaces and tools that stay out of the way.

SKILLS

Backend: Laravel, NestJS, Node.js, Symfony

Data: Apache Kafka, Elasticsearch, MongoDB, MySQL, PostgreSQL, Redis

Infrastructure: AWS, Docker, GitHub Actions, Kubernetes

API: GraphQL, gRPC, OpenAPI

Language: JavaScript, PHP, Swift, TypeScript

Frontend: CSS, HTML, React, Tailwind CSS

WORK EXPERIENCE

Chief Technology Officer

[Safety Real Time](#) — Remote

Apr 2022 - Present

(4 years 7 months)

Directed technical strategy for a cloud-based fleet management platform serving enterprise trucking operators. Owned full-stack architecture: Node.js, NestJS, and Redis, covering pre-trip inspection workflows, accident and damage reporting, driver scorecards, live operational visibility, and compliance document management for logistics clients.

- **Fleet management platform architecture** — Defined the full-stack cloud architecture for a fleet management SaaS, from mobile driver apps through backend ingestion services to live operations dashboards used by enterprise fleet managers.
- **Real-time event pipeline** — Engineered high-frequency data ingestion for live vehicle and driver events, with Redis pub/sub propagation maintaining sub-second freshness across operations dashboards.
- **Safety event routing** — Designed time-sensitive routing for safety-critical incidents: pre-trip failures, accident reports, and driver threshold breaches, reaching frontline operators with minimal propagation delay.
- **Compliance and document workflows** — Built scheduling, reminder, and document management modules enabling fleet managers to track regulatory renewals, inspection dates, driver scorecards, and maintenance compliance.
- **Engineering team leadership** — Led cross-functional delivery across backend, frontend, and infrastructure disciplines aligned to the product roadmap and SLA commitments with enterprise clients.

Technology: Node.js, NestJS, TypeScript, Redis, React, PostgreSQL, Elasticsearch, Docker, Kubernetes, AWS, GitHub Actions

Senior Software Engineer

[167Pluto](#) — Belgrade, Serbia

Feb 2024 - Jun 2025

(1 year 5 months)

Led backend architecture for a high-concurrency online betting platform. Delivered fault-tolerant, event-driven microservices on Node.js and NestJS with zero-downtime deployments through peak sporting events.

- **Event-driven domain services** — Designed NestJS and Kafka flows across bet lifecycle, odds distribution, and settlement contexts.
- **Idempotent command handling** — Introduced explicit idempotency keys enabling full replay safety across the settlement pipeline.
- **Cache partitioning strategy** — Protected hot shards and preserved read performance consistently during peak traffic windows.
- **CI/CD hardening** — Added staged rollouts and automated canary analysis to enable zero-downtime production deployments.
- **Observability baseline** — Instrumented tracing, structured logging, and latency dashboards to shorten incident diagnosis time during live events.

Technology: **Node.js, NestJS, TypeScript, Apache Kafka, Redis, PostgreSQL, Elasticsearch, Docker, Kubernetes, GitHub Actions, GraphQL**

Software Engineer

[Apoddo](#) — Remote

Jan 2021 - Mar 2022

(1 year 3 months)

Designed and delivered the BFF layer for Humanity.com, an enterprise workforce management platform with hundreds of thousands of users. Aggregated multi-service calls into lean, UI-optimized payloads via Node.js and NestJS, with Redis caching and OpenAPI contracts decoupling frontend and backend release cycles.

- **BFF layer design** — Aggregated 3–5 upstream API calls into single, UI-optimized payloads, removing redundant round-trips and decoupling frontend rendering from backend schemas.
- **Redis cache-aside pattern** — Added TTL-based invalidation for high-frequency schedule endpoints, cutting backend load and improving response consistency across the enterprise user base.
- **OpenAPI-documented contracts** — Decoupled frontend and backend release cycles through stable, versioned API contracts, enabling independent delivery velocity across teams.
- **Graceful degradation design** — Implemented circuit breaker patterns to serve cached or partial responses when upstream Humanity.com services were slow or unavailable, preserving frontend availability under backend instability.
- **Serialization profiling** — Optimized response serialization paths on data-heavy workforce management screens, reducing latency under real production traffic.

Technology: **Node.js, NestJS, TypeScript, Redis, PostgreSQL, GraphQL, OpenAPI, Docker, AWS**

Software Engineer

[Evermed](#) — Remote

Jan 2021 - Aug 2021

(8 months)

Led end-to-end migration of legacy PHP/Symfony monoliths to AWS microservices. Designed automated media processing pipelines for a healthcare education platform with secure, traceable video transcoding.

- **Monolith migration** — Decomposed a PHP/Symfony codebase into independently deployable AWS services with zero delivery gaps.
- **Transcoding automation** — Built end-to-end pipelines with idempotent retries and traceable state transitions at every stage.
- **Control-plane separation** — Isolated orchestration logic from compute-heavy media workers for independent, targeted scaling.
- **Audit logging** — Added lifecycle logging across the media asset flow for full compliance traceability end to end.
- **Infrastructure as code** — Codified AWS environment provisioning so rollouts, rollbacks, and recovery remained predictable and reviewable.

Technology: **PHP, Symfony, AWS, Docker, JavaScript, MySQL, GitHub Actions**

Lead Back-end Developer

[Pannovate](#) — Remote

Jan 2020 - Dec 2020

(1 year)

Led backend across two generations of LaunchPad, maintaining the Symfony 4 core-banking monolith while architecting Sylo, a full NestJS microservice rewrite. Mentored the team through the migration and established engineering standards as the platform scaled.

- **Sylo microservice architecture** — Modelled and built Sylo's NestJS service layer from scratch, defining bounded contexts, typed gRPC and Protocol Buffer contracts, and event-driven flows via Kafka and Redis.
- **Secure-by-default patterns** — Enforced OAuth2 least-privilege boundaries and secure-by-default communication across every internal service boundary.
- **Kubernetes deployment topology** — Built Kubernetes deployment flows for Sylo services, tuned for resilience, rolling updates, and fast pod-level recovery at production scale.
- **Backend team mentorship** — Tutored backend engineers on Node.js, NestJS, TypeORM, and event-driven paradigms during the transition away from the Symfony monolith.
- **Transactional correctness safeguards** — Added double-entry style balance checks and idempotent handlers across money-moving service boundaries to prevent inconsistency under concurrent load.

Technology: **Node.js, NestJS, TypeScript, PostgreSQL, Redis, Apache Kafka, gRPC, Symfony, Docker, Kubernetes, AWS, GitHub Actions**

Back-end Developer

[Quantox Technology](#) — Belgrade, Serbia

Nov 2018 - Jan 2020

(1 year 3 months)

Backend engineer outsourced through Quantox to Catena Media on AskGamblers.com, an iGaming affiliate platform with 150,000+ registered users. Refactored legacy Symfony modules achieving resource usage reductions of up to 90%, and hardened performance of SEO-critical content delivery pipelines under sustained traffic load.

- **Legacy refactoring** — Decomposed multiple legacy Symfony modules into cleaner service boundaries, achieving up to 90% resource usage reductions on high-load paths and resolving issues that had gone unaddressed for months or longer.
- **SEO-critical delivery paths** — Hardened response times and cache strategies on SEO-sensitive public endpoints, sustaining performance under hundreds of thousands of monthly visits.
- **Database and cache tuning** — Profiled and resolved PostgreSQL and Redis bottlenecks in data-intensive review and bonus workflows running under sustained production traffic.
- **Affiliate tracking integrations** — Wired server-to-server tracking and commission reporting integrations with upstream operator APIs under strict regulatory constraints.
- **Review aggregation services** — Maintained and extended backend services powering user-submitted casino reviews, moderation queues, and reputation scoring for 150,000+ registered users.

Technology: **PHP, Symfony, MongoDB, Redis, MySQL, Elasticsearch, JavaScript, HTML, CSS**

Back-end Developer

[Spark451](#) — Remote

Sep 2017 - Jun 2018

(10 months)

Built modules and features on Element451, a higher-education CRM handling hundreds of thousands of student admissions. Most notable contribution: architecting Intelligent Admissions inside Decision451, which auto-evaluated each application by processing hundreds of candidate properties to generate scores and trigger the appropriate enrollment workflows.

- **Intelligent Admissions engine** — Modelled and implemented the Intelligent Admissions module inside Decision451, auto-evaluating admissions by processing hundreds of per-application properties, generating scores, and applying configured enrollment workflows automatically.
- **Admissions scoring pipelines** — Built configurable, rule-based scoring pipelines in PHP/Laravel handling hundreds of thousands of student admissions across 20+ partner institutions.
- **Reporting query optimization** — Refactored and optimized reporting queries against large institutional datasets to sustain throughput during peak enrollment windows.
- **Scheduled batch jobs** — Delivered reliable daily analytics exports to institutional decision-makers across concurrent enrollment cycles without missed runs.
- **Legacy refactoring** — Refactored legacy PHP code and contributed to planning and development of new CRM modules, improving maintainability across the platform.

Technology: **PHP, Laravel, MongoDB, JavaScript, AWS, Docker**

Back-end Developer

[CrelTive](#) — Belgrade, Serbia

Apr 2016 - Sep 2017

(1 year 6 months)

Developed and maintained multiple backends for websites, web applications, APIs, and e-commerce systems across concurrent agency client projects. Contributed to an in-house deployment workflow and helped mentor colleagues, while delivering PHP (Laravel and Symfony), Node.js, and JavaScript implementations across a range of client engagements.

- **Multi-stack backend delivery** — Shipped backend implementations in PHP (Laravel and Symfony) and Node.js across multiple concurrent client engagements, adapting to varied project requirements without quality drop.
- **Reusable backend modules** — Built shared libraries covering recurring agency patterns: form handling, CMS scaffolding, API clients, eliminating duplicate effort across projects.
- **Build and deploy automation** — Contributed to the agency's in-house custom deployment workflow and tooling, removing manual steps from release flows and reducing human-introduced errors.
- **Project scaffolding** — Standardized repeatable development environments and project templates across the agency's active portfolio for faster project kickoff.
- **Colleague mentorship** — Helped and mentored junior engineers on PHP patterns, deployment practices, and code quality to improve overall team output and delivery consistency.

Technology: **PHP, Laravel, Symfony, Node.js, JavaScript, MySQL, HTML, CSS**

Full Stack Web Developer

Freelance — Belgrade, Serbia

Feb 2015 - Apr 2016

(1 year 3 months)

Delivered full-stack web solutions for clients in Serbia, including a bespoke CMS built from scratch in raw object-oriented PHP with AJAX-driven JavaScript, designed to give multiple clients a centralized content management surface without off-the-shelf overhead.

- **Bespoke CMS from scratch** — Modelled and built a custom CMS in raw object-oriented PHP with AJAX and JavaScript, giving clients a centralized, maintainable admin interface tailored to their specific content workflows.
- **Full-stack delivery** — Handled database design through frontend UI across every engagement, with no handoffs or external dependencies at any layer.
- **Lean architecture decisions** — Scoped backend designs to the actual problem: purposeful, maintainable, and built with room to grow without pre-emptive complexity.
- **Performance tuning** — Improved page load and API responsiveness through targeted server-side caching, query optimization, and asset delivery improvements.
- **End-to-end client engagement** — Managed requirements gathering, iterative delivery, and post-launch support directly with each client from first brief through production.

Technology: **PHP, Laravel, JavaScript, MySQL, HTML, CSS**

EDUCATION

Bachelor's Degree in Software and Information Engineering

[Singidunum University](#) — Belgrade, Serbia

Oct 2016 - Oct 2020

(4 years 1 month)

Bachelor of Software Engineering. Curriculum focused on distributed systems, cloud computing, DevOps, and AI/ML with a hands-on, project-oriented capstone track.

High School Diploma in Mathematics and Computer Science

[Mathematical Grammar School](#) — Belgrade, Serbia

Sep 2011 - Jun 2015

(3 years 10 months)

Graduated from a High National Distinction institution specializing in advanced mathematics, physics, and computer science. Completed a full-stack Electronic School Diary as a graduation project.

LANGUAGES

English

Full professional proficiency

Serbian

Native speaker